

1. Temporal locality refers to this property exhibited by most algorithms.
recently referenced variables will be referenced again soon
2. Variables referenced in the body of these constructs will surely exhibit high temporal locality
for loops
nested loops
while loops
3. After not finding a datum in the _____, the _____ will try to load it from _____.
cache
hardware
main memory
4. When a CPU attempts to fetch an instruction or a data operand, the memory system will look first in the _____ level.
fastest
5. Variables referenced in the body of these constructs will surely exhibit high spatial locality.
None of these answers
6. After not finding a datum in the cache, the system will look next into the _____.
main memory
7. Data transfers between registers and cache are managed by the _____.
programmer
compiler
8. Spatial locality refers to this property exhibited by most algorithms.
variables close (in the memory) to recently referenced variables will be referenced soon
9. On modern computer systems, the memory system is organized as an _____, rather than a flat system.
hierarchy
10. When a CPU attempts to fetch an instruction or a data operand, the memory system will look first in the _____.
cache
11. Caching consists of copying information into a more expensive storage system; a hard drive can be viewed as a cache for the network clouds.
more expensive
a hard drive
12. Instructions referenced in the body of these constructs will surely exhibit high spatial locality.
sequential code with no loops
while loops
for loops
nested loops
13. The level with the smallest storage capacity is assigned the number 1.
Registers - 1

Cache - 2

Main Memory - 3

Hard drive - 4

Network storage - 5

14. Data transfers between main memory and cache are managed by
hardware

15. Caching consists of copying information into a _____ storage system; _____ can be viewed as
a cache for the hard drive.

faster

main memory